

MENG-465

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Design Project 2

Heat Sink

I used the geometry given by the textbook. This included 3 fins that run in-line with the flow of air through the inlet.

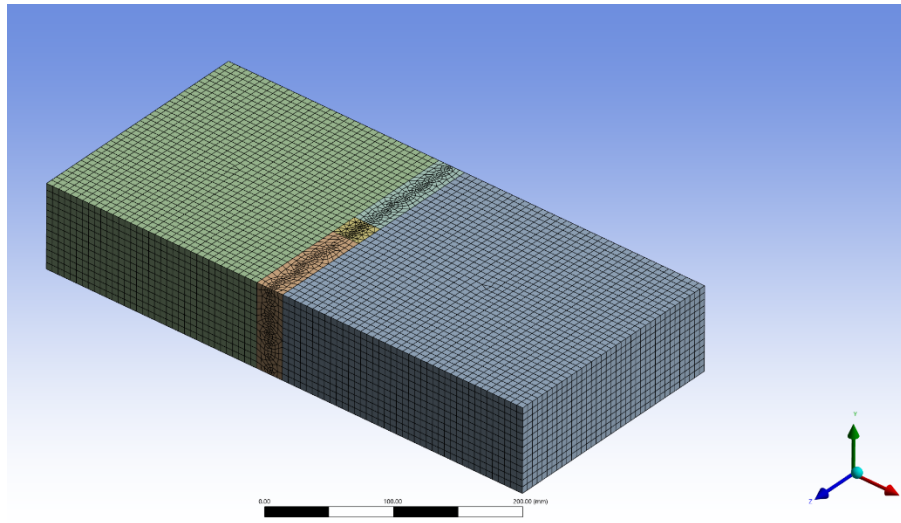


Figure 1. Full mesh of heat sink and domain.

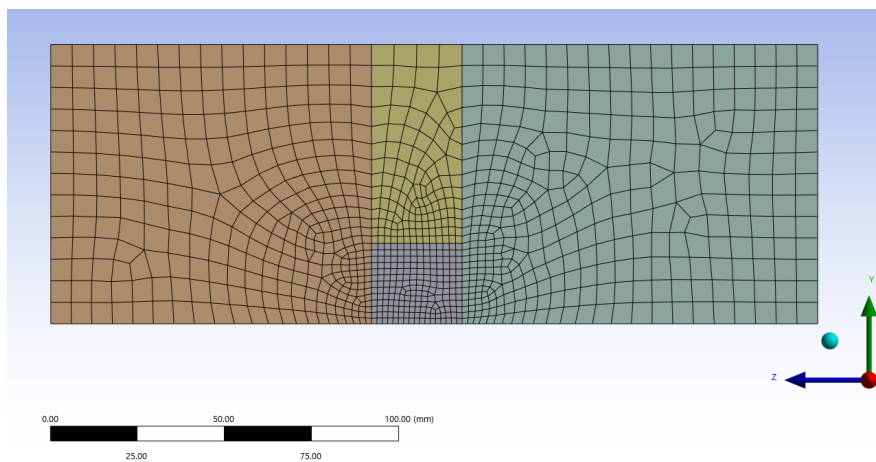


Figure 2. Cross sectional showing mesh of domain immediately around the heat sink.

Statistics	
Nodes	78140
Elements	155700
Show Detailed Statistics	No

Figure 3. Mesh Statistics

I meshed the system as 6 domain segments and 1 heatsink. The up-and-downstream Segments consisted larger elements. Ideally, I would bias the mesh to have smaller components towards the middle of the mesh (as seen in the textbook), but I was running into element number problems. Thankfully, I was able to increase my element number from 50,000 to ~156,000. The mesh on the actual heatsink contains smaller elements as to reduce interpolation error between nodes. Because of this, the Fluent contour has a more accurate depiction of heat dissipation through the heat sink.

For materials, I stuck with aluminum for the heat sink. Based on what I found online, aluminum is a relatively common and cheap option for a heat sink. It has decent thermal performance, and is cheaper than copper. The domain is air, which would be an accurate fluid/domain if this was inside a computer or the like. As for boundary conditions (BC), the velocity inlet could be simulation for a fan blowing air in its direction, and the gauge pressure being 0 is because we simulate it at our atmosphere.

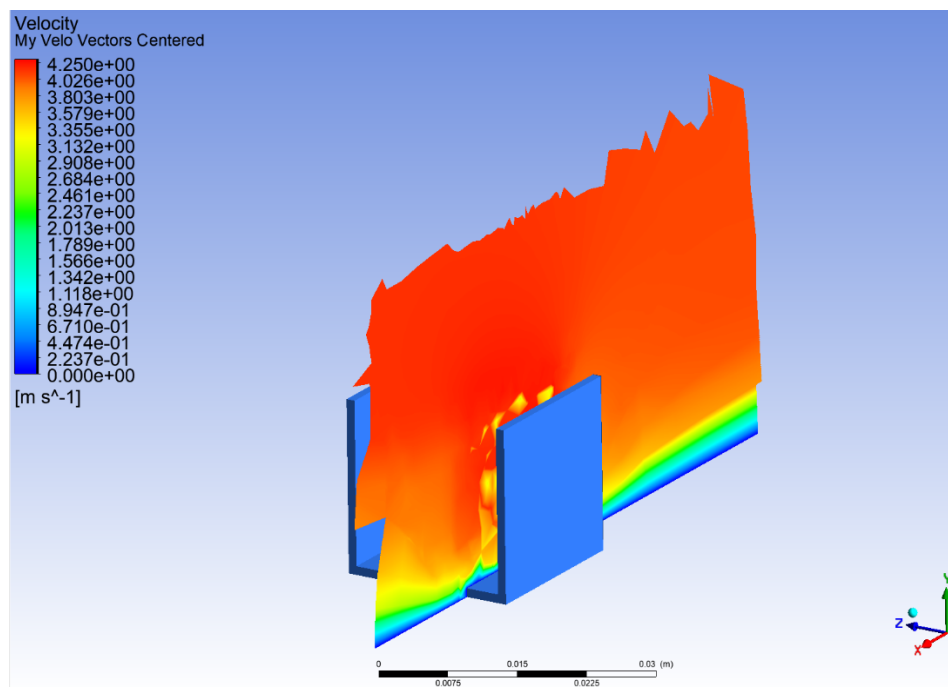


Figure 3. Velocity vectors of air between two fin blades.

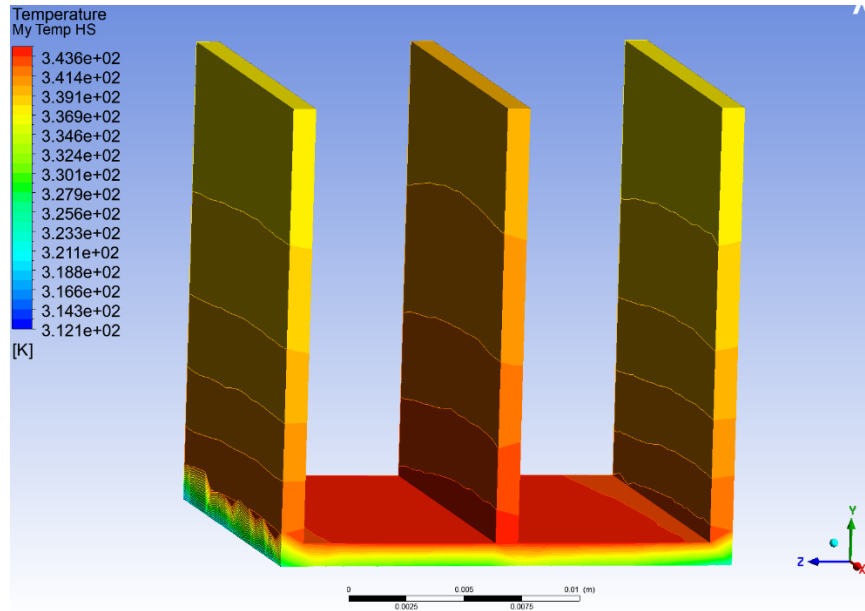


Figure 4. Temperature contours on surface of heat sink.

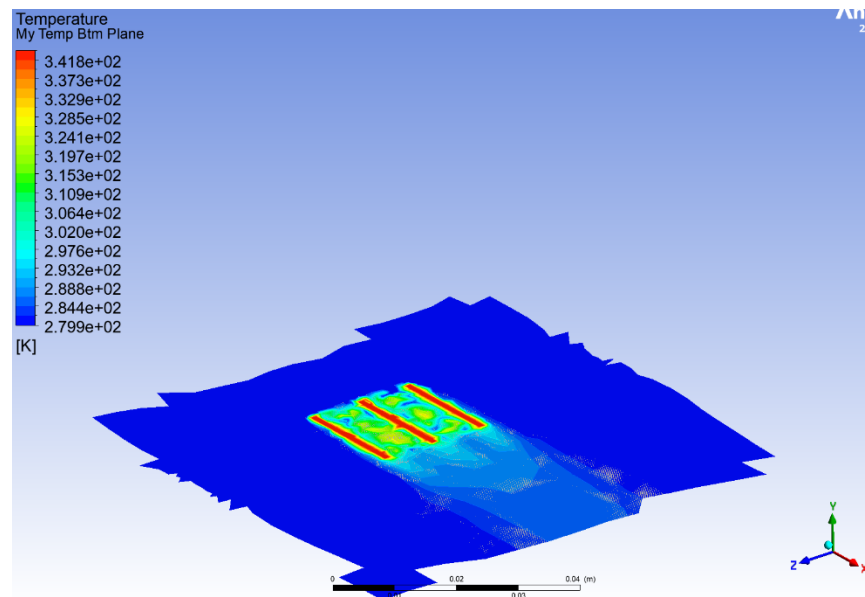


Figure 5. Temperature contours on base of heat sink (in case this is what you meant by surface).